

Procedure: Attach an IAM Role to an EC2 Instance (S3 access without keys)

Step 1 — Create Role in Amazon Web Services IAM

1. Open IAM Console → Roles → Create role
2. Trusted entity: Select AWS Service
3. Use case: Choose EC2
4. Click Next

Step 2 — Add Permissions

1. Search policy: AmazonS3FullAccess
2. Select it → Next

Step 3 — Name the Role

- Name: EC2-S3-Role
- Click Create role

Step 4 — Attach Role to EC2 Instance

1. Go to EC2 Dashboard
2. Select running instance
3. Actions → Security → Modify IAM role
4. Select EC2-S3-Role → Update

Step 5 — Verify from Instance (No Access Keys!)

SSH into instance and run:

```
aws s3 ls
```

```
aws ec2 describe-instances
```

- `aws s3 ls` Works
- `aws ec2 describe-instances` **Unauthorized (no EC2 permission)**

Reason: Instance receives **temporary credentials** from IAM Role automatically.

scenario-based questions on IAM Roles, Users, Groups, and Policies in Amazon Web Services.

1. A new developer joins your team and needs access to EC2 and S3. How will you grant permissions securely?
2. A user requires access to AWS resources for only two days. How can you provide temporary access?
3. A developer accidentally exposed AWS access keys on GitHub. What immediate steps will you take?
4. A user wants to access AWS services using the AWS CLI. How will you enable programmatic access?
5. Fifty employees require the same S3 permissions. How will you manage this efficiently?
6. You must remove access permissions for 20 users at once. What is the easiest method?
7. HR, Developers, and Admin teams need different levels of access. How will you organize permissions?
8. An EC2 instance must access S3 without storing access keys. What is the correct approach?
9. Account A needs to access resources in Account B. How can cross-account access be configured?
10. A Lambda function must read and write data to DynamoDB. How will you assign permissions?
11. A user needs administrator privileges for a short time. How can you grant this safely?
12. A user should access only one specific S3 bucket. How do you restrict access?
13. Users should be prevented from deleting any AWS resources. How can you enforce this?
14. A user should be allowed to log in only during office hours. How can this be implemented?
15. Access must be allowed only from the office network IP address. How can this be controlled?
16. What are the best practices to manage IAM permissions securely?